Dubai International Academy - Al Barsha

Grade 10 MYP Physics Criteria B & C Summative Assessment (Mr. Pathan)

IB Middle Years
Physics

Grade: Grade 10 MYP

Criteria: B 17 C 21

Total: 38 points

Duration: 50 mins

Close time: Oct 09, 11:32 am (School TZ)

Aarav (Aarav) Pai

Submitted on Oct 08, 11:42 am (School TZ)

Score: 0/38 pts

Criterion	Criteria with points	Equivalent achievement level
B-Inquiring and designing	0/17	-/8
C-Processing and evaluating	0/21	-/8

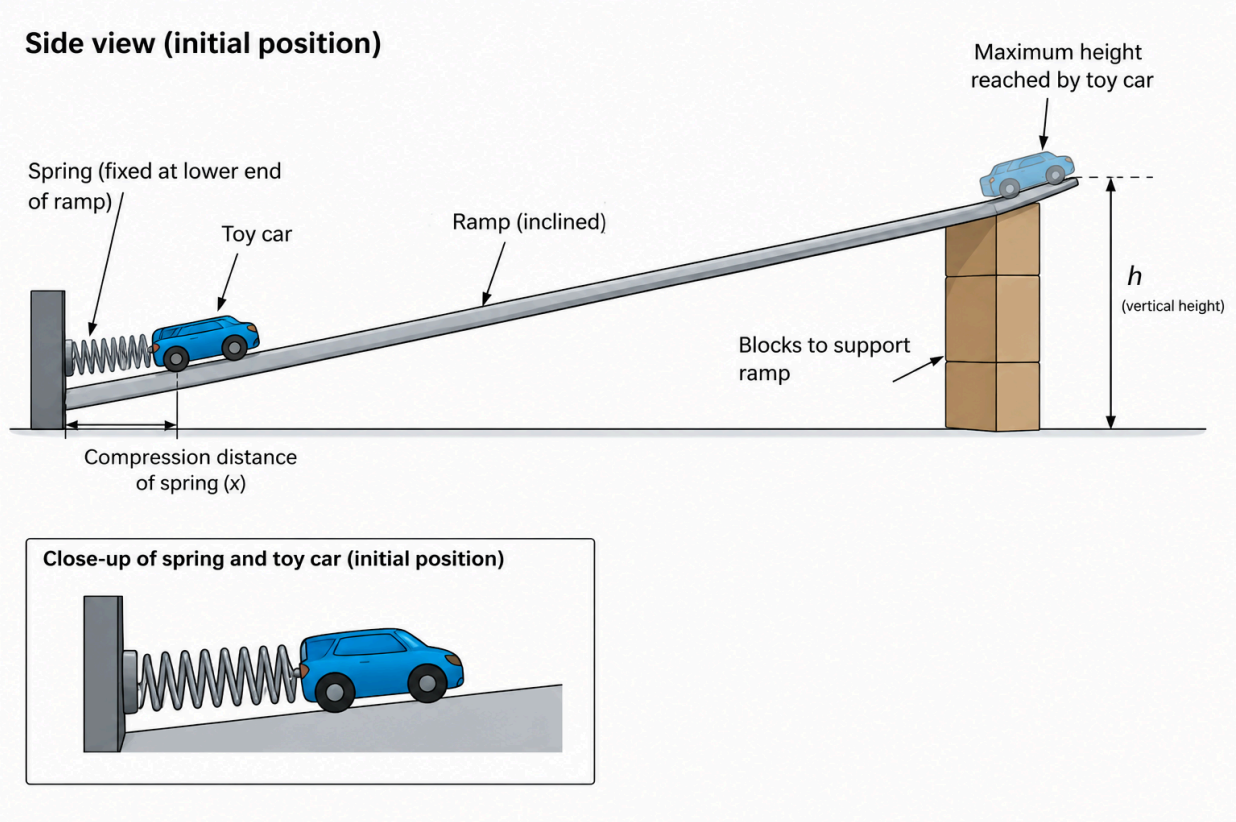
Experiment on Conservation of Energy

A group of students wants to investigate the transfer of energy from elastic potential energy to gravitational potential energy.

The students have the following equipment:

- a toy car
- a spring
- an inclined ramp
- a protractor (not shown in the diagram)
- blocks to support the ramp

The spring is fixed at the lower end of the ramp. The toy car is placed against the spring and the spring is compressed by a measured distance. When released, the spring pushes the car up the ramp. The car eventually stops at its maximum height.



Q 1.1 **Design** an investigation to determine the relationship between the compression of the spring and the maximum gravitational potential energy gained by the toy car.

Your investigation design should include:

- research question **[1]**
- independent and dependent variables **[1]**

- two relevant controlled variables and justification for controlling these variables **[4]**
- additional two equipment required to carry out the investigation **[2]**
- a clear and detailed procedure **[5]**
- a description of the data to be collected for the investigation to be reliable. **[3]**

Student's answer

Research question:
How does changing the Compression (cm) (2,4,6,8,10) of the spring affect the maximum gravitational potential energy (J) while keeping the hieght of the spring constant.
Idependant variables:
The comrpession of the spring cm (2,4,6,8,10)

Dependant variable:
The maximum gravitation potential energy achived by the car.

Control Variables:

1. The Mass of the car: Changing the mass of the car can change the gravitantional potential energy as the gpe=mgh, where changing m can skew the results. We must weigh the car before satrting the experiment.
2. Height of the ramp. Changing the hieght of the ramp can change the maximum height of the toy car. This should be controlled so that the data is not sweked.

Additional Equipment:

1. A Weighing scale to measure the mass of the car.
2. A meter ruler to find the height of the ramp

Procedure:

- Toy car
- A spring
- inclined ramp
- a protractor

- blocks to support the ramp
- A meter ruler
- Weighing scale.

1. Start by setting up equipment as shown in the diagram.
2. Measure The height of the ramp
3. Weigh the toy car and note donw the mass.
4. Now place the car on the spring and pull it back 2 cm
5. Release the car and observe. Measure the maximum hieght and note it down in the data table.
6. Repeat this process two more times for a total of three trials.
7. Now pull the car back to 4 cm and repeat of three trials
8. Repeat the steps for three trials, note the data down and do the same for the rest of the IV group heights.

We must have 3 trials so that we can stop the anomalies
Find the average of all the results for the 5 groups and write them down in a new processed data table.

Markscheme

	1 mark	2 marks	3 marks	4 marks	5 marks
RQ	States a question involving **compression distance of spring** and **maximum gravitational energy attained by the car**.				
IV & DV	Identifies the independent (compression distance) **AND** dependent variable (maximum height).				
CVs	Identifies **one relevant controlled variable** without justification.	Identifies **two relevant controlled variables** without	Identifies one relevant controlled variable and gives a relevant justification for **one**		

			controlled variable.	Identifies two relevant controlled variables and gives relevant justifications for **both** controlled variables.	
Eq	Identifies suitable additional equipment needed to measure **height**, e.g., ruler	Identifies suitable additional equipment needed to **height**, e.g., ruler, **AND** mass, e.g., digital weighing scale OR balance			

P	Describes how the spring is compressed consistently at the first initial compression distance, while controlling CVs.	Describes how to create and **measure different compression distance**.	Describes how the car is **released consistently** and how its maximum height is measured (from a reference point).	Describes repeating the procedure for **different compression distances** while controlling other variables.	Gives a **complete, logical procedure**, including conducting multiple trials **AND** calculating the average speed.
D	States that **at least 5 different compression distance** should be tested and measurements repeated.	States that **at least 5 different compression distance** should be tested and measurements repeated **AND** specifies a suitable range of compression and **at least three trials per height**.	States that **at least 5 different compression distance** should be tested and measurements repeated **AND** specifies a suitable range of compression and **at least three trials per height.t**, with a mean/average calculated.		

Teacher's comments B -

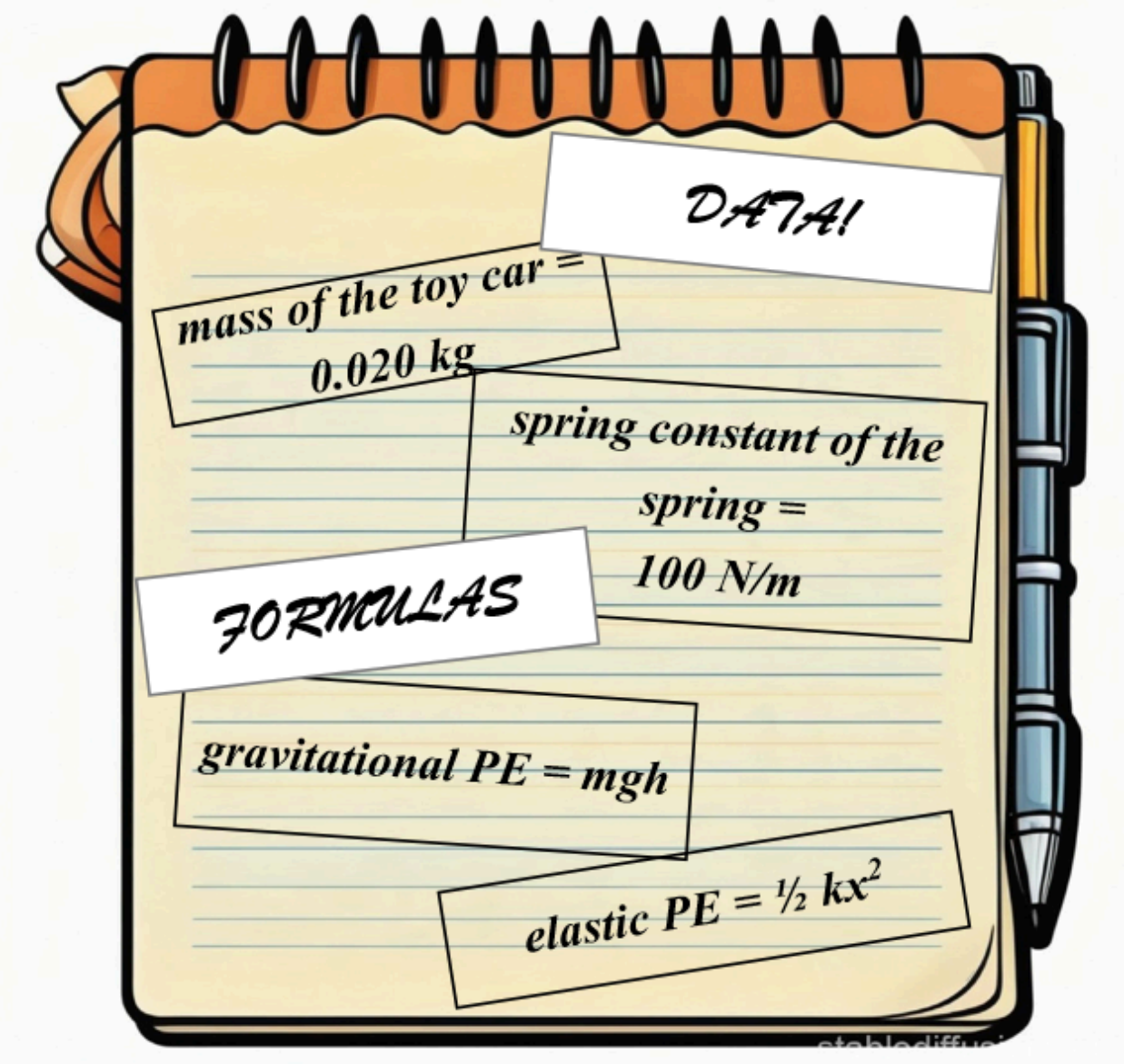
No comments for this question

Another group carried out the investigation to determine the elastic potential energy stored in the spring and the gravitational potential energy stored in the toy car at the end of the ramp.

Their results are shown below.

Compression of spring / m	Maximum vertical height gained / m
0.04	0.031
0.06	0.071
0.08	0.122
0.10	0.199
0.12	0.296

The following data are also obtained by the group, together with formulas for the elastic potential energy and gravitational potential energy:



The group formulates a hypothesis as:

"The elastic potential energy stored in the spring is not equal to the gravitational potential energy achieved by the toy car when it reaches the maximum height because some of the mechanical energy will be lost as it moves up the ramp."

Q 1.2 **State** a conclusion about whether the results support the group's hypothesis. **Use** evidence from the data obtained by the group to justify your conclusion.

Student's answer

The hypothesis is partially supported as the data of the compression of the spring and data of the height of the toy car are not proportional.

Markscheme

1 mark — The results support the hypothesis

1 mark — Evidence for the energy relationship: Uses data to show that GPE on top is not equal to EPE of the spring.
For example, at 0.04 m of compression distance, EPE = 0.08 J while GPE = 0.04 J.
Therefore, the claim that the GPE is equal to the KE is not supported.
Do not accept the first mark without a sample working.

Teacher's comments C -

No comments for this question

Q 1.3 A member of the group suggested that there are a number of sources of systematic error in the experimental setup that could explain their results.

Identify one major source of systematic error from the experimental setup. **Explain** its impact on the results of the experiment.

Student's answer

The maximum height of the toy car could me a value that can be read wrong as the car can roll back.

Markscheme

1 mark — **The cart did not start from the very bottom of the ramp**
1 mark — meaning its initial gravitational potential energy was not zero/was not accounted for *owtte*.

Award max 1 mark for ANY logical systematic error from the setup.
For example,
The ramp was not perfectly smooth, so friction acted on the cart as it travelled up the ramp, causing some energy to be transferred to the surroundings.

Teacher's comments C -

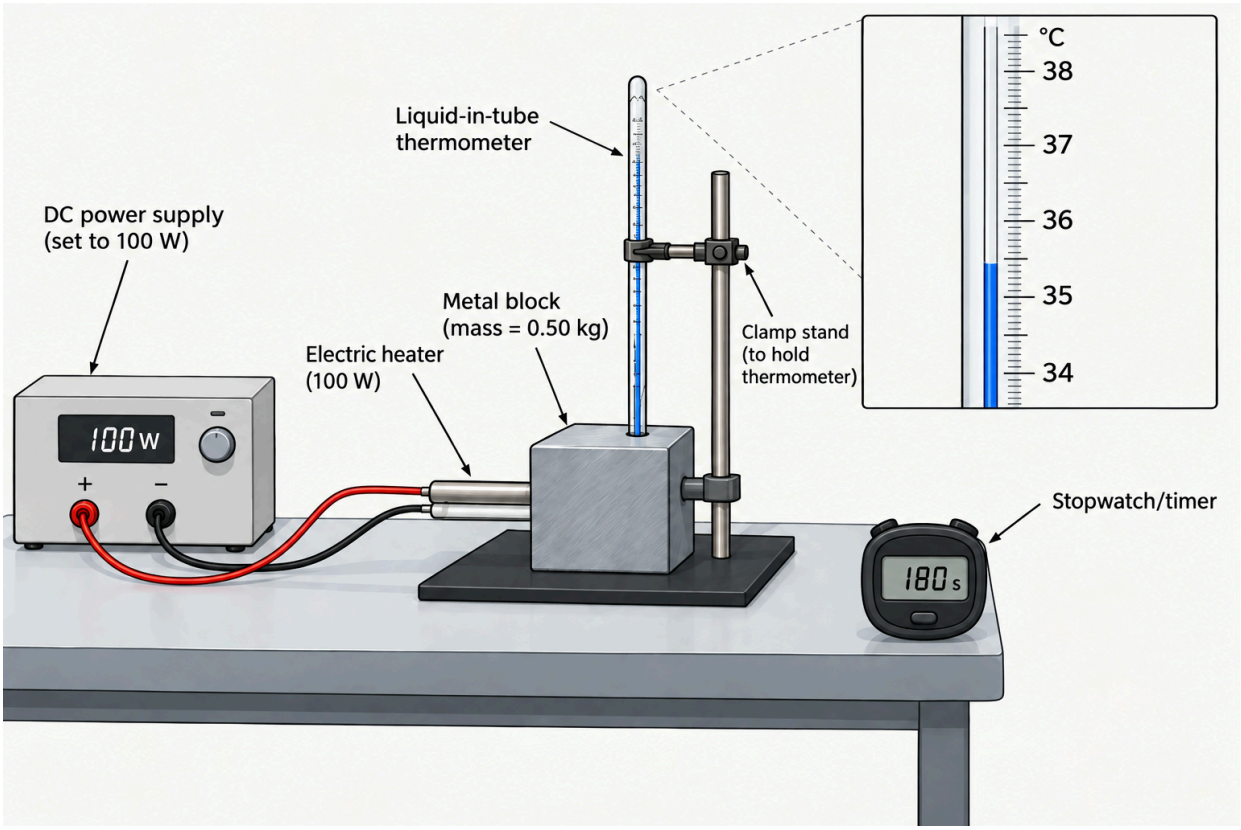
No comments for this question

Experiment on Specific Heat Capacity

A group of students investigates the specific heat capacity of a metal block. They use an electric heater to transfer energy to the block and measure its temperature at regular time intervals.

The mass of the metal block is **0.50 kg**. The heater has a power of **100 W**.

The experimental setup is shown with the metal block is uninsulated as it lays on the base of the black metal retort stand.



Q 2.1 **State** one safety consideration that should be followed when carrying out this investigation.

Student's answer

The electric heater transfers heat to the metal block so the students should wear heat resistant gloves when touching the block or this can cause burns.

Markscheme

Any valid safety consideration. Examples include, but not limited to:

- Use heat-resistant gloves or tongs when handling the heated metal block.
- Do not touch the electric heater while it is switched on.
- Allow the metal block to cool before handling it.

- Keep electrical connections away from water or other liquids.
- Switch off the power supply before adjusting or removing the apparatus.
- Handle the thermometer carefully to avoid breaking it.

Teacher's comments B -

No comments for this question

Q 2.2 The temperature is measured every 1 minute.

The temperature at t = 180 s is shown in the diagram.

Measure the temperature at t = 180s. **Complete** the table, including the column heading and the missing time when the temperature is 43.9 degree Celsius.

Time(s)	Temperature(c)
0	22.0
60	26.5
120	31.0
180	34.5
240	39.7
300	43.9
360	48.0

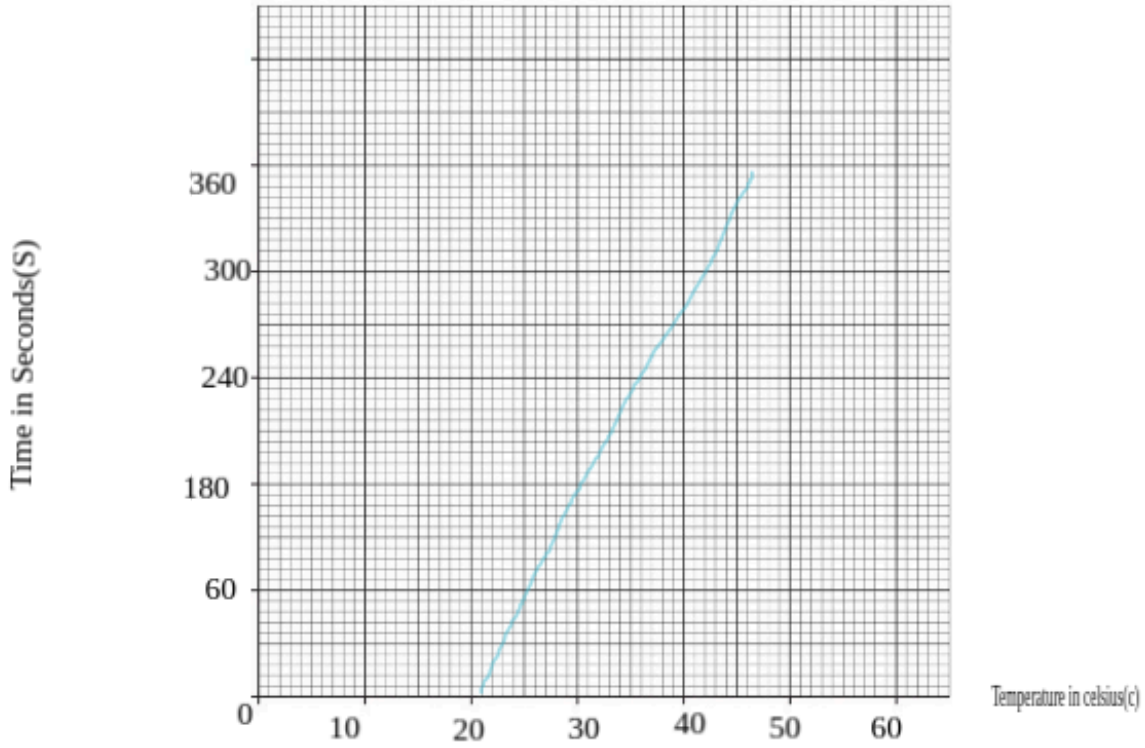
Markscheme

1 mark — time in first column heading AND temperature in second column heading
1 mark — both units correct; second for time AND degree Celsius for temperature
1 mark — 35.4 (temperature at t = 180 s)
1 mark — 300 (time when T = 43.9 degree Celsius)

Teacher's comments C -

No comments for this question

Q 2.3 **Plot** a line graph to show the relationship between time of heating and temperature of the metal block. Start your graph at (0,0).



Markscheme

- **1 mark** — Time of heating end plotted on the x-axis and temperature on the y-axis, both with units
- **1 mark** — Appropriate, even scales are used and occupy a reasonable proportion of the graph, i.e, at least half of the grid
- **1 mark** — All six points are plotted accurately (plus/minus 1 small square)
- **1 mark** — Suitable line of best fit is drawn.

Teacher's comments C -

No comments for this question

Q 2.4 **Show that** the rate of change in temperature $\frac{\Delta T}{\Delta t}$ is about 0.07 degree Celsius per second.

Student's answer

360-0/48-22

Markscheme

1 mark — states that rate of change in temperature $\frac{\Delta T}{\Delta t}$ = gradient of the line

1 mark — shows calculation of gradient of line; two points must be at least half of the line apart.

Teacher's comments C -

No comments for this question

Q 2.5 The relationship between the power *P* supplied to a substance, its mass *m*, its specific heat capacity *c*, and the rate of change in temperature $\frac{\Delta T}{\Delta t}$ is given by:

$$\mathbf{P = m \times c \times \frac{\Delta T}{\Delta t}}$$

where ***P*** is power, ***m*** is mass, and ***c*** is specific heat capacity.

Calculate the specific heat capacity of the metal using this equation. Include its unit.

Student's answer

100=0.5xcx0.07
c=100/0.5x0.07
c=2857.1 Kg/k/j

Markscheme

1 mark — Correctly rearranges the equation: **c = P ÷ (m × rate of change in temperature)**

1 mark — Correct answer: **c = 2857 J kg^{-1} °C^{-1} ≈ 2900 J kg^{-1} °C^{-1}** or **2.9 × 10^{3} J kg^{-1} °C^{-1}**

Teacher's comments C -

No comments for this question

Q 2.6 The student suggested that the calculated specific heat capacity of the metal block in Q2.5 is less than the actual specific heat capacity of the metal block.

Explain this suggestion.

Student's answer

Student didn't write an answer

Markscheme

1 mark — States that the metal block is uninsulated OR that some of the energy supplied by the heater is transferred to the surroundings rather than the metal block
1 mark — Explains that this makes the measured rate of temperature increase **smaller**, so using **P = m × c × rate of change in temperature** would give a calculated specific heat capacity **greater than**, not less than, the actual value.

Teacher's comments C 0 / 2

No comments for this question

Q 2.7 The student wants to extend the investigation a with a new pair of independent variables to determine the specific heat capacity of a metal block.

The new independent variable is the electric power of the source, and the new dependent variable is temperature of the heating, while controlling the duration of heating

Formulate a scientific hypothesis for this new set of variables.

IF:	THEN:	BECAUSE:
If the electric power of the source increases	Then the temperature of the block will also increase	Because the energy will be transferred

Markscheme

IF-THEN
1 mark —
IF: States that increasing the **electric power** increases the temperature of the metal block
THEN: Predicts that the **temperature of the metal block will be higher** after the same fixed duration of heating.

BECAUSE:
2 marks —
CORRECT BECAUSE Statement: Explains that greater power transfers **more energy per unit time** to the metal block, resulting in a greater temperature

increase; award only 1 mark if logical BECAUSE statement but not stated with scientific terminologies.

Max 1 mark —

For an incorrect BECAUSE Statement but logically and scientifically related justification/explanation.

Teacher's comments C -

No comments for this question